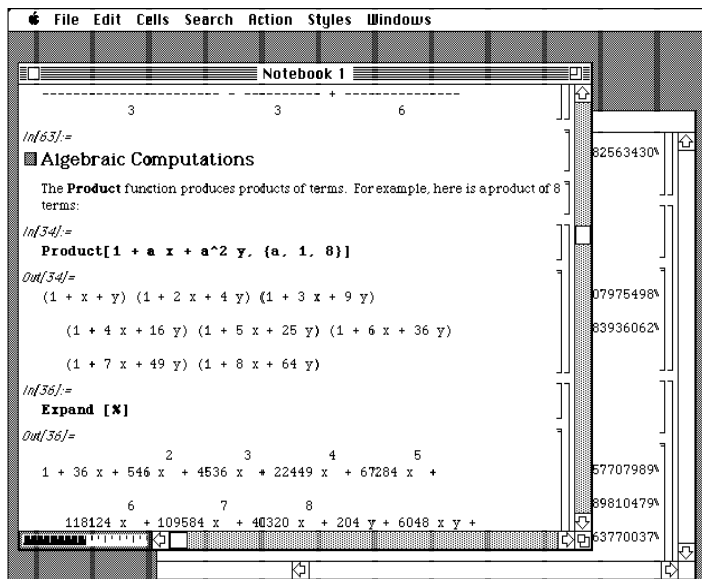


■ 1.3.2 Special Topic: Notebooks

More advanced front ends for *Mathematica* often support objects called “notebooks”.

A notebook is a file that contains text and graphics, mixed with *Mathematica* input and output.

Here is a *Mathematica* notebook on a Macintosh in which text has been inserted to document lines in a *Mathematica* session. On a Macintosh, you can choose a variety of different fonts and styles for your text.



One application of notebooks is simply as a way of organizing and documenting your *Mathematica* calculations.

Another important way to use notebooks is as computerized analogs of textbooks. You can put in a notebook the same kind of text and graphics that would appear in a standard textbook. In addition, however, you can include *Mathematica* input that sets up the calculations described in the text. In this way, you can use the notebook as something like a “live” textbook. You can read the text to learn about a topic, and then execute the associated *Mathematica* input, to actually perform a *Mathematica* calculation.

Web sample page from The *Mathematica* Book, First Edition, by Stephen Wolfram, published by Addison-Wesley Publishing Company (hardcover ISBN 0-201-19334-5; softcover ISBN 0-201-19330-2). To order *Mathematica* or this book contact Wolfram Research: info@wolfram.com; <http://www.wolfram.com/>; 1-800-441-6284.

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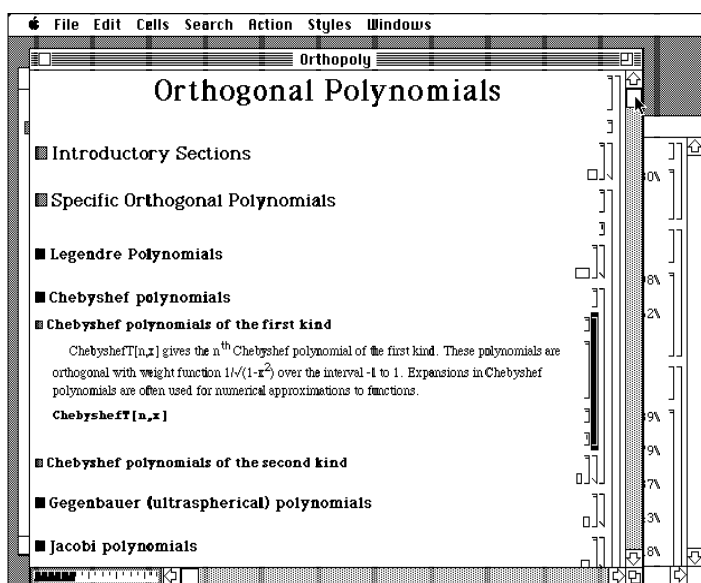
Here is a notebook about orthogonal polynomials. The notebook contains text about orthogonal polynomials, together with *Mathematica* definitions that allow you to do calculations with orthogonal polynomials.



Mathematica notebooks are typically broken up into a sequence of “cells”. Each cell can contain one piece of *Mathematica* input or output, or a paragraph of text, or a piece of graphics. When you edit a notebook, you typically operate on complete cells in much the same way as you operate on individual characters of text.

In many *Mathematica* notebooks, it is convenient to organize the cells into a hierarchy of “hypercells”. Each section of a notebook, for example, may be a single hypercell, containing a sequence of many individual cells. *Mathematica* front ends typically allow you to “close” hypercells, so that only their headings are visible. If you see a closed hypercell whose heading seems relevant, you can typically use graphical tools to “open” the hypercell, and see the details of the cells inside it.

The notebook on orthogonal polynomials is organized into a hierarchy of “hypercells”. This shows the hypercells corresponding to sections in the notebook. One hypercell is “open”, showing the next of cells and hypercells inside it.



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